

Scientific Figure Captioning with Multimodal LLMs & Knowledge Graphs

Motivation

Scientific figures contain complex information that is often challenging for Large Language Models (LLMs) to grasp from visuals and captions alone. This can lead to generic or factually incorrect descriptions (hallucinations). This project aims to solve this by creating a system where a Multimodal LLM leverages a Knowledge Graph (KG) to generate captions that are more accurate, detailed, and factually grounded. The core idea is that by providing the LLM with verified facts, we can significantly improve its scientific reasoning capabilities.

Research Question & Objectives

Does augmenting a Multimodal LLM with a Knowledge Graph improve the factual accuracy and semantic richness of generated captions for scientific figures?

Objectives

Build a data pipeline that extracts figures and metadata from scientific papers.

Construct a multimodal Knowledge Graph from textual and visual information.

Implement a retrieval-augmented captioning system that uses the KG to provide context to the LLM.

To evaluate and quantify the performance of the KG-augmented model against a baseline.

Dataset & Setup

We use the ArXivCap dataset which is made up of 6.4M images and 3.9M captions from 572K research papers across scientific domains. Dataset structure.

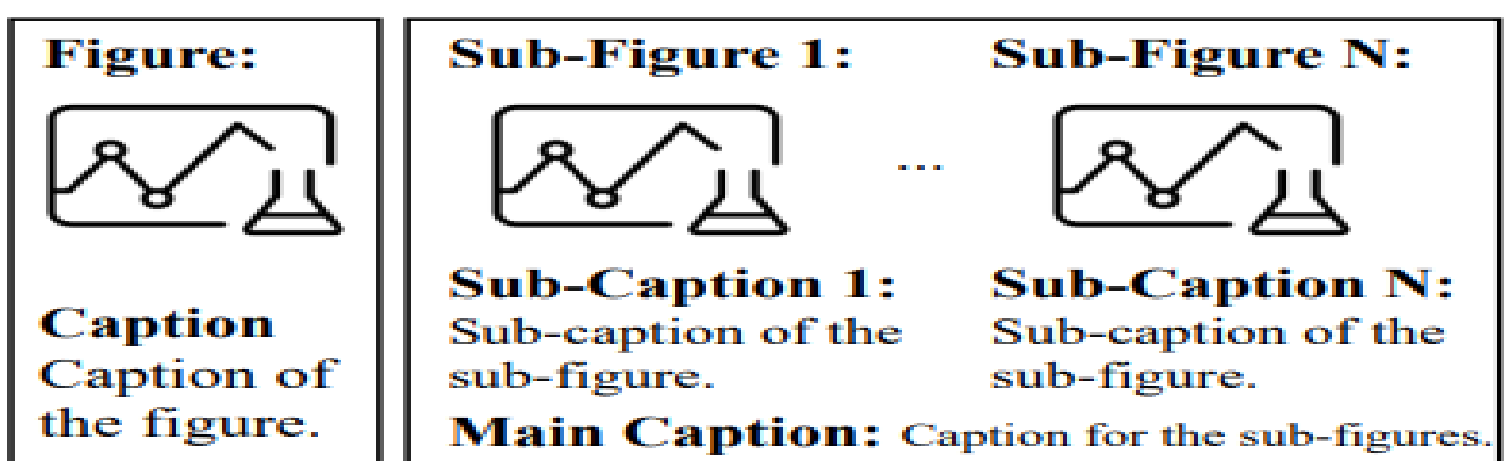


Figure 1: ArXivCap figure/sub-figure layout. Source: ArXivCap (MMInstruction).

Each paper contains figures; a figure may have sub-figures with their own sub-captions and a main caption for the whole figure (see insert).

Our working set is a physics-focused subset for the main run (diverse plots, spectra, diagrams) to keep the KG coherent. We also produce small out-of-distribution (OOD) slice (Computer Science) to probe robustness when the KG has little or no coverage.

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Methodology

We work on a **physics-focused subset of ArXivCap** (plots, spectra, schematic diagrams). Each record includes the **figure image**, the **human caption**, and paper **metadata**. Using **Gemini in text-only mode**, we read the *human caption* and extract concise **triples** (*subject, relation, object*) such as (*[Fe/H], CORRELATES_WITH, stellar mass*). Surface forms are normalized (lowercase, punctuation trimming), obvious chart synonyms are folded (*line plot* → *line graph*), and relations are canonicalized to a small controlled set (*MEASURES, CORRELATES_WITH, INCREASES, DECREASES, CONSISTENT_WITH, DISCOVERED_BY, ...*). Using **Gemini in vision-only mode**, we analyze the *image* and produce up to **10 visual tags** as short noun phrases (*line graph, spectrum, error bars, x-axis, legend*). We load a compact schema into **Neo4j**: nodes **Figure, Entity, VisualTag**; edges **Figure→MENTIONS→Entity, Figure→HAS_TAG→VisualTag, VisualTag→IS_VISUAL_CUE_OF→Entity**. Entity and tag strings are deduplicated (normalization + fuzzy matching), and per-figure uniqueness constraints keep the graph clean.

For a given figure, we retrieve **only its own subgraph** (no cross-paper or cross-figure hops). Concretely, we pull the **1-hop neighborhood** around the **Figure** node: all **MENTIONS** edges to **Entity** nodes, all **HAS_TAG** edges to **VisualTag** nodes, and any **IS_VISUAL_CUE_OF** links that connect those tags back to entities. This keeps context grounded in what the image shows. We then **rank** these facts with a simple visual-aware score: (i) **tag overlap** (facts touching entities that are pointed to by the figure's current **VisualTag** set score higher), (ii) **entity centrality** within the figure subgraph (variables that connect to more relations get a small boost), and (iii) **relation priority** (informative scientific links like **MEASURES/CORRELATES_WITH/INCREASES/DECREASES** outrank generic ones). We keep **top-K ≈ 8–12** high-precision facts. Using **Gemini in multimodal mode (image + text context)**, we generate a **draft caption (C₀)**, ≤ ~35 words, with instructions to describe the figure concisely using only the supplied facts and what is visually present. **Gold captions are never used as context** (no leakage).

From **C₀**, **Gemini (text-only)** extracts its own claims back into **triples**. We **verify** those triples against the same figure-local KG: **supported** (edge exists or normalized equivalent) are kept; **contradicted** (conflicts with an existing edge) are dropped; **neutral** (no evidence either way) are retained cautiously. We add a **tiny enrichment** (strict 1-hop) around **verified** entities e.g., a unit, a named method, or a directly linked variable re-ranked with the same visual-aware score and capped to keep prompts crisp. Finally, **Gemini (image + verified/enriched facts)** regenerates the **final caption (C₁)**. This is the caption we report and evaluate against a **zero-shot baseline** (Gemini on **image-mode only**).

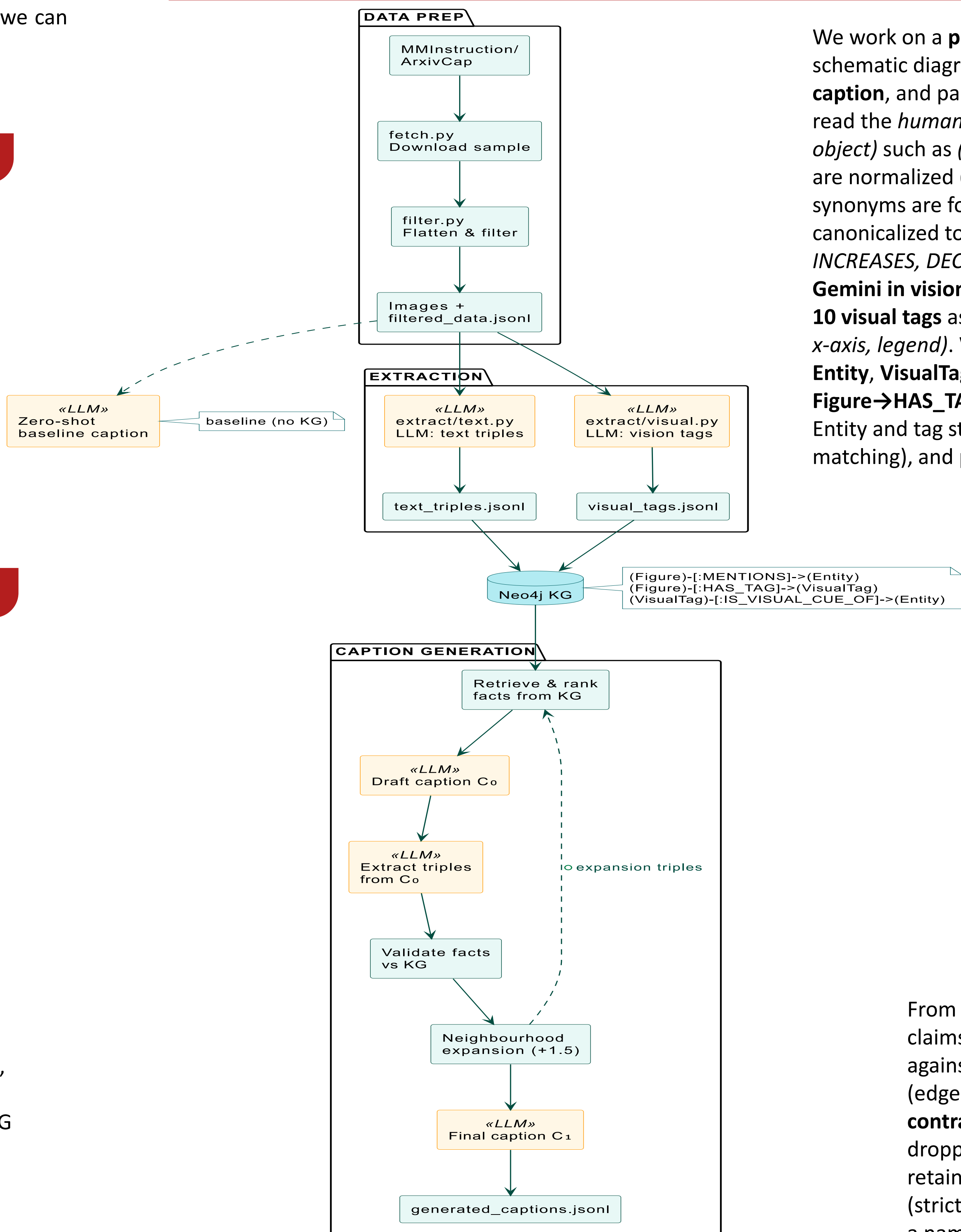


Figure 2. Pipeline: sample figures → extract text/vision facts → build Neo4j KG → retrieve local facts → draft, verify, enrich → final caption

Results

Entity Precision / Recall / F1: compare entities mentioned in the generated caption against entities from the gold caption (entities are obtained by the same extractor used to build the KG, with the same normalization).
Grounding accuracy: share of entities in the generated caption that can be **unambiguously grounded** to a node in the figure's KG subgraph (1 if the mention maps to exactly one node after normalization; 0 otherwise).

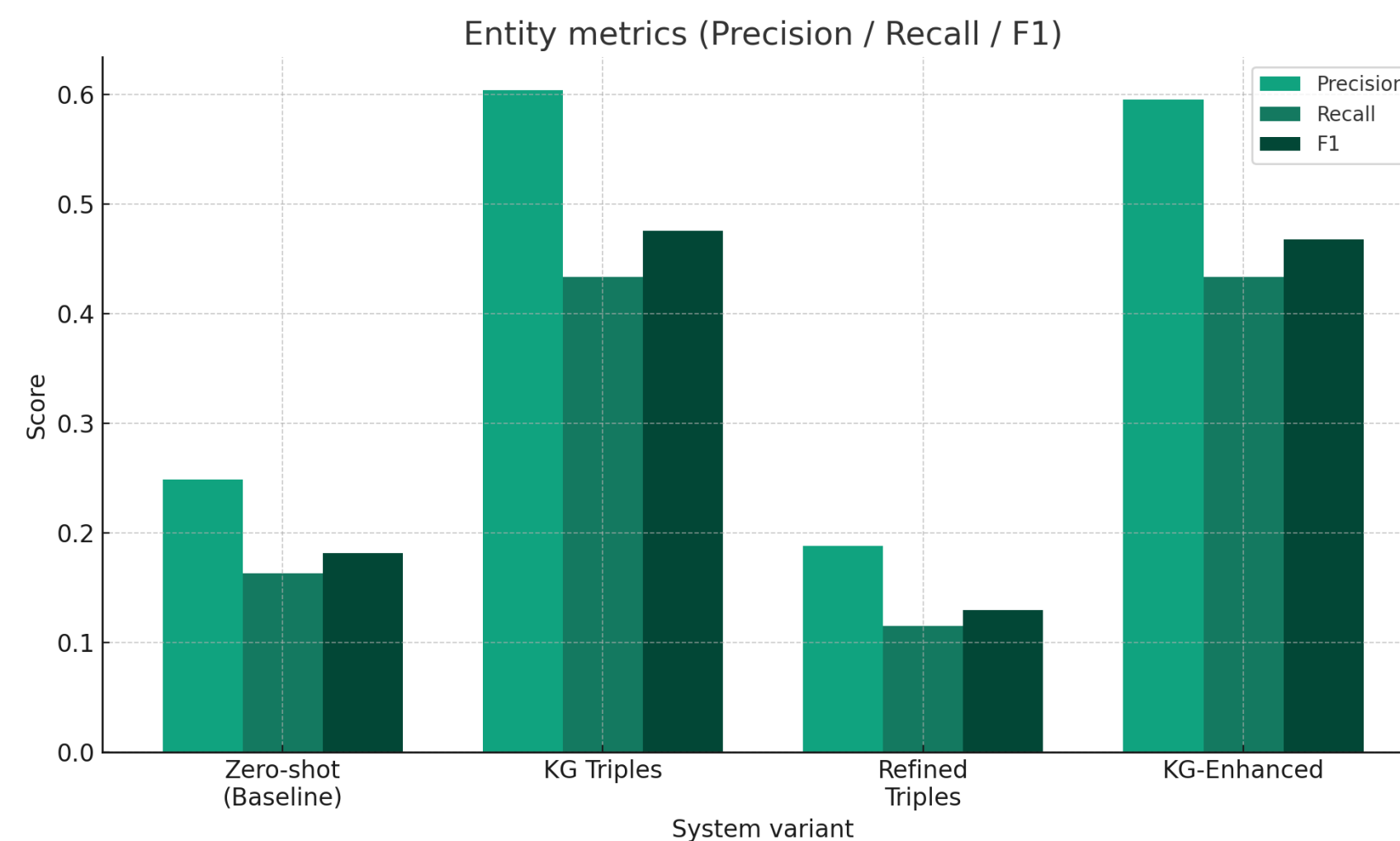


Figure 3 — Entity metrics (Precision / Recall / F1).

Bars compare four system variants using the **same multimodal LLM**: *Zero-shot* (image only), *KG Triples* (image + ranked 1-hop KG facts), *Refined Triples* (regenerate from verified model claims only), and *KG-Enhanced* (verified claims + small 1-hop enrichment). **KG Triples** and **KG-Enhanced** deliver ~2.6× higher F1 than zero-shot, driven by much higher precision; **Refined Triples** loses recall by discarding retrieval facts.

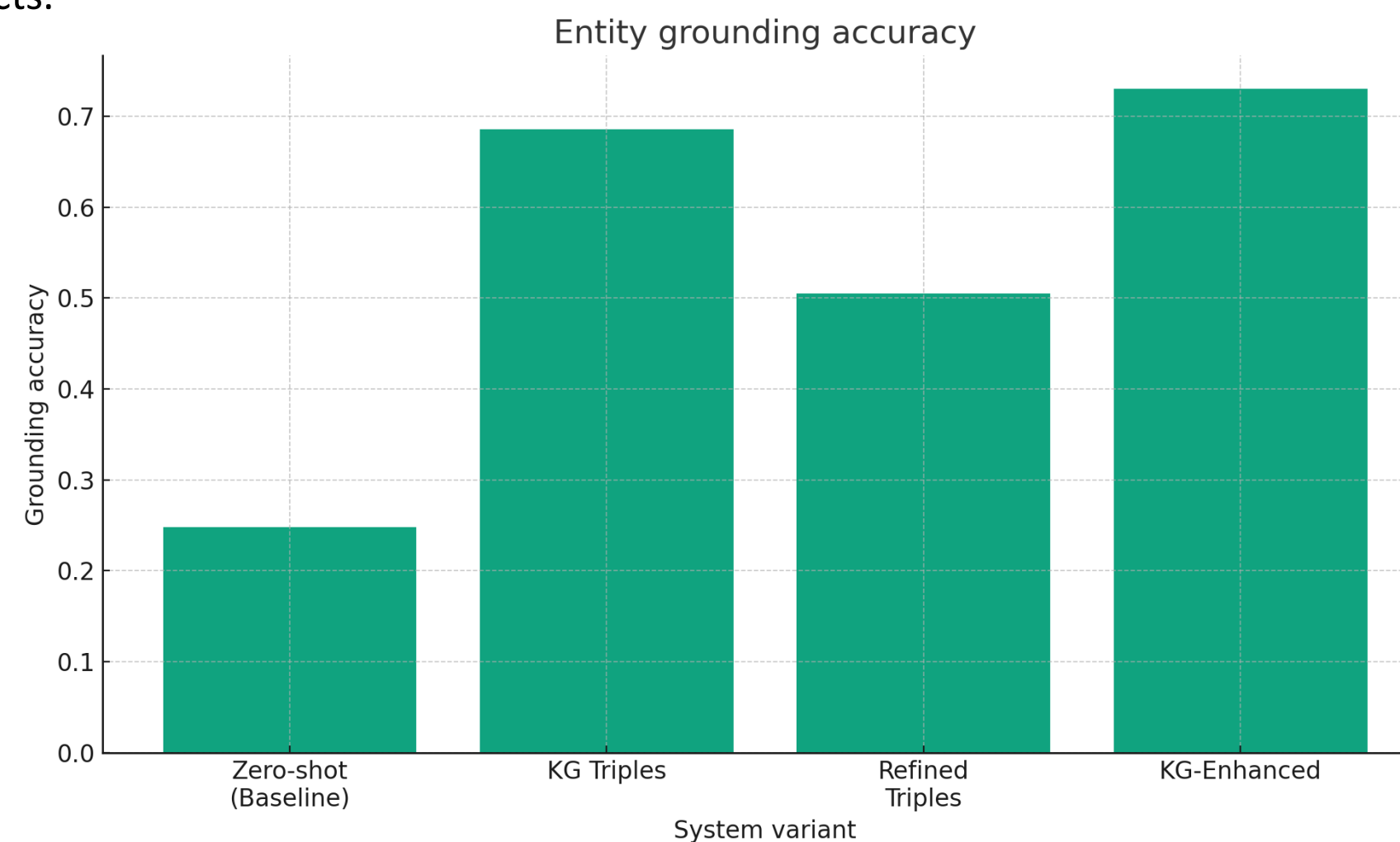


Figure 4 — Entity grounding accuracy.

Accuracy of mapping caption entities to **nodes in the same figure's KG subgraph**. **KG-Enhanced** is best (~0.73), outperforming *KG Triples* (~0.68) and far above **Zero-shot** (~0.25). The small enrichment step helps the model name entities that actually exist in the graph, improving factual grounding.

Conclusion & Future work

On a physics-centric scientific KG, conditioning a **multimodal LLM** on **figure-local verbalised triples** raised entity **F1 ~0.18→~0.47** and **grounding ~0.25→~0.73 (~3×)** versus image-only; a **refined-only** pass (regenerating from verified claims) cut recall, whereas **verify-then-enrich** with a tiny **1-hop** slice recovered recall while preserving precision, yielding the most grounded captions. These gains hinge on **KG construction quality** a physics-aware schema (Figure, Entity, VisualTag), strict normalization/deduplication, and tight links from caption text and visual tags that keep retrieval precise. **In future work** we will integrate an **LLM-as-judge** signal for automated fact-checking within the verification loop and build a **larger, better-constructed scientific KG** (more figures, stronger normalization/identifier hygiene) to further improve retrieval and grounding.

References

Zhang, T., Kishore, V., Wu, F., Weinberger, K., Artzi, Y. *BERTScore: Evaluating Text Generation with BERT*. ICLR 2020.
*Xu, Z. et al. *Retrieval-Augmented Generation with Knowledge Graphs*. 2024.
*Li, L. et al. *Multimodal ArXiv: A Dataset for Improving Scientific Comprehension of LLMs (ArXivCap & ArXivQA)*. ACL 2024. [6.4M images, 3.9M captions from 572k papers].